

FRAME FOR PERFORMANCE

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-461 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

SENSOR RESEARCH MISSION

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-462 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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INSTRUMENT THE DESIGN

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GM-463 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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CALIBRATE FOR PERFORMANCE

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GM-464 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

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SENSOR RESEARCH MISSION

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GM-465 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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VALIDATE THE DESIGN

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GM-466 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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EVALUATE FOR PERFORMANCE

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GM-467 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

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SENSOR RESEARCH MISSION

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GM-468 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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STRESS-TEST THE DESIGN

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GM-469 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

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GM-470 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.